

0518

10.

	Yes	No	Total
Back ↑	187 (166)	113 (134)	300
College	162 (166)	138 (134)	300
High ↓	149 (166)	151 (134)	300
Total	498	402	900

$$300 \times \frac{498}{900} = 166$$

$$300 \times \frac{402}{900} = 134$$

H_0 : marriage rate and educational levels 無關 (Dependent)

H_a : 有關 (Independent)

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}} = \sum \frac{O_{ij}^2}{E_{ij}} - n$$

$$= \frac{187^2}{166} + \frac{162^2}{166} + \frac{149^2}{166} + \frac{113^2}{134} + \frac{138^2}{134} + \frac{151^2}{134} - 900 = 10.06$$

$$\chi_{3, 0.01}^2 = 6.25 < 10.06 \Rightarrow \text{Reject } H_0$$

$\Rightarrow$ marriage rate and educational levels 有顯著關連

$\alpha = 0.01$ H_0 : 三種 residence 的家庭人口分布無顯著差異
 H_a : 有顯著差異

12.

	Apartment	Duplex	Single	Total
1	8 (9.67)	20 (9.67)	1 (9.67)	29
2	16 (11)	8 (11)	9 (11)	33
3	10 (11.34)	10 (11.34)	14 (11.34)	34
4 ↑	6 (8)	2 (8)	16 (8)	24
Total	40	40	40	120

$$40 \times \frac{29}{120} = 9.67$$

$$40 \times \frac{33}{120} = 11$$

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{O_{ij}^2}{E_{ij}} - n = \frac{8^2}{9.67} + \frac{20^2}{9.67} + \frac{1^2}{9.67} + \frac{16^2}{11} + \frac{8^2}{11} + \frac{9^2}{11} + \frac{10^2}{11.34} + \frac{10^2}{11.34} + \frac{14^2}{11.34} + \frac{6^2}{8} + \frac{2^2}{8} + \frac{16^2}{8} - 120$$

$$= 36.46$$

$$\chi_{6, 0.01}^2 = 16.81 < 36.46 \Rightarrow \text{Reject } H_0$$

$\Rightarrow$ 三種 residence 之間的家庭人口分布有顯著差異

	Apartment	Duplex	Single
1	0.2	0.5	0.025
2	0.4	0.2	0.225
3	0.25	0.25	0.35
4 ↑	0.15	0.05	0.4

12.	A	4.8	5.2	5.0	4.9	5.1				
	B	5.0	4.7	4.9	4.8	4.9				
	B	A	B	A	B	B	B	A	A	A
a.	4.7	4.8	4.8	4.9	4.9	4.9	5.0	5.0	5.1	5.2
Rank	1	2.5	2.5	5	5	5	7.5	7.5	9	10

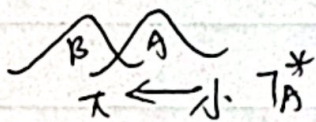
$$T_A = 2.5 + 5 + 7.5 + 9 + 10 = 34 \quad T_A^* = 5(5+5+1) - 34 = 21$$

$$T_B = 1 + 2.5 + 5 + 5 + 7.5 = 21$$

H_0 : 上游溶氧量 = 下游溶氧量 H_a : 上游 > 下游

$$T = \min(34, 21) = 21 > 19 \Rightarrow \text{Non-Reject } H_0$$

$\Rightarrow$ 沒有足夠證據顯示該城鎮排放污水導致溶氧量顯著下降



b. $H_0: \mu_1 = \mu_2$, $H_a: \mu_1 > \mu_2$

$$\bar{x}_A = 5, \quad s_A^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{0.1^2 + 0.2^2 + 0.1^2 + 0.1^2}{4} = 0.025$$

$$\bar{x}_B = 4.86, \quad s_B^2 = \frac{0.14^2 + 0.16^2 + 0.07^2 + 0.06^2 + 0.04^2}{4} = 0.013$$

$$s_p^2 = \frac{4 \times 0.025 + 4 \times 0.013}{5+5-2} = 0.019$$

$$t = \frac{5 - 4.86}{\sqrt{0.019 \left(\frac{1}{5} + \frac{1}{5} \right)}} = 1.606$$

$$t_{8, 0.05} = 1.806 > 1.606 \Rightarrow \text{non-reject } H_0$$

10(a)

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13. H_0 : Eye-movement rate between deaf and hearing children are same
 H_a : Eye-movement rate differ between deaf and hearing children

$$T_1 = 15 + 11 + 18 + 10 + 14 + 12 + 17 + 16 + 13 = 126$$

$$T_1^* = 9(9+9+1) - 126 = 45$$

$$T = \min[45, 126] = 45$$

$n_1 = 9, n_2 = 9, \alpha = 0.05$. Critical value = 68. $> 45 \Rightarrow$ Reject H_0
 $\Rightarrow$ 在 $\alpha = 0.05$ 下, 有充分證據顯示 Eye-movement differ between deaf and hearing children

4. + + + - + + +

H_0 : 兩母體分布相同. H_a : 兩母體分布不同, $\alpha \leq 0.1$

$$X = 6$$

$$X \sim \text{Bin}(7, 0.5)$$

$$P(X=0) = C_0^7 \times 0.5^7 = 0.0078$$

$$P(X=7) = C_7^7 \times 0.5^7 = 0.0078$$

$$P(X \leq 1) + P(X \geq 6) = 0.0078 + 0.0078 = 0.0156 < 0.1$$

$$P(X \leq 0) + P(X \geq 7) = 0.0078 + 0.0078 = 0.0156 < 0.1$$

Rejection Region $\Rightarrow X=0$ or $X=7 \Rightarrow$ 沒有 $X=6$

$\Rightarrow$ Non-Reject H_0